Strict Endpoint Optimization for the Physical TPC Route:

A Nonduplicated Loss Registry, Exact Rational Primal–Dual Certificates, and the Equality Stop

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Abstract

TPC-MVP1 requires every physical reconstruction loss to be charged exactly once and the final fixed-power loss to satisfy the *strict* inequality

$$\Lambda_{\mathrm{phys}} < \frac{1}{400}.$$

Recent papers isolate the literal costs of physical-frame reassembly, residual grouping, fixed-shift localization, complete tail return and TPC-18 full-block cover. Their existence does not yet imply endpoint compatibility: the costs may overlap, may be unknown on the growing carrier, or may exhaust the whole reserve.

We define a canonical amplitude-loss registry. Each physical operator event receives one provenance identifier and one energy-to-amplitude convention. Exact duplicates are merged; distinct events are added; a joint theorem replaces its constituent tokens rather than being charged in addition to them. On each fixed route and affine scale cell, endpoint optimization becomes the rational linear program

$$\Lambda_* = \min\{c^\mathsf{T} x + d : Ax \geq b,\ x \geq 0\}.$$

Its dual is

$$\max\{b^\mathsf{T} y + d : A^\mathsf{T} y \leq c,\ y \geq 0\}.$$

An exact feasible primal point below $1/400$ is a strict compatibility certificate. An exact dual point at or above $1/400$ is a stop certificate only for the same complete actual theorem-backed registry. Matching rational primal and dual objectives equal to $1/400$ prove that equality is unavoidable and therefore *does not* pass.

Applying the registry to the present TPC interfaces yields a deliberately incomplete verdict: the growing exponents for several frame, grouping, localization, tail and cover tokens are not proved, so assigning them value zero would be invalid. The paper supplies the exact endpoint decision mechanism and a rerouting rule, but no strict physical certificate for the actual fixed-shift carrier, no new L2 estimate and no twin-prime theorem.

Keywords: strict endpoint; physical loss ledger; linear programming; dual certificate; provenance; stop–go audit.

Ledger convention. All tokens below are amplitude exponents. If an inherited theorem is stated for squared norms, its exponent is converted once before entry. The determinant–zero reserve and the H5 compatibility reserve are separate ledgers and may not be spent here.

1 Why strict optimization is a theorem gate

TPC-112 proves that determinant–zero compatibility and the physical endpoint are independent unless a map-specific intertwining theorem is supplied [2]. TPC-113 identifies the canonical physical-frame cost; TPC-114 the residual grouping/native-Gram cost; TPC-115 the exact fixed-h_0 localization cost; TPC-116 the complete tail return; and TPC-117 the full-block cover/remainder interface [3–7]. Each is a necessary input, not a free estimate.

Suppose an upstream raw packet has the fixed reserve

$$X^{1-\frac{1}{400}+o(1)}.$$

If physical transport costs $X^{\Lambda_{\mathrm{phys}}+o(1)}$, a positive fixed-power margin remains only when

$$\boxed{\Lambda_{\mathrm{phys}} < \frac{1}{400}.}$$

At equality the result is $X^{1+o(1)}$ at best. The unspecified $o(1)$ cannot be assigned a favorable sign, so equality is a hard stop for this route.

2 Canonical nonduplicated loss tokens

Definition 2.1 (Physical loss token)**.** A token is a record

$$\tau = (\mathrm{id}, \mathrm{map}, \mathrm{scope}, \mathrm{norm}, \lambda, \mathrm{source}, \mathrm{deps}),$$

where:

(i) id names the literal operator event;

(ii) map includes domain, codomain and normalization;

(iii) scope lists the actual active labels and fixed shift;

(iv) norm records whether the source estimate was an amplitude or energy estimate;

(v) $\lambda \geq 0$ is the converted amplitude exponent;

(vi) source is the theorem supplying the bound; and

(vii) deps lists constituent events already absorbed by a joint bound.

An unknown λ is denoted ?, not 0.

Definition 2.2 (Canonical registry)**.** For one route, first convert every theorem to an amplitude exponent. Merge records only when their identifiers, literal maps, scopes and normalizations are identical. If a joint token τ_J is used, delete precisely the constituent tokens listed in $\mathrm{deps}(\tau_J)$. The remaining identifiers form the canonical registry $\mathcal{T}$, and

$$\Lambda_{\mathrm{phys}} = \sum_{\tau \in \mathcal{T}} \lambda_\tau.$$

This sum describes one literal multiplicative loss chain. Parallel branches, mutually exclusive routes and alternative joint factorizations have separate registries and are optimized routewise; they are not added unless a theorem identifies them as factors in the same physical composition. When different theorems bound the same physical event, they are alternative evidence, not additional events: a declared route selects one theorem-backed bound (or a proved joint improvement) before summing the registry.

Proposition 2.3 (No-double-charge alternatives)**.** *Assume every physical operator event in a route occurs in exactly one retained registry token, either individually or as a declared dependency of one joint token. Then Λ_{phys} charges every event once. In contrast:*

(a) retaining a joint token and one of its dependencies double-charges that event;

(b) merging two equal numerical exponents with different maps can omit a real loss;

(c) entering an unknown token as zero asserts an unproved $X^{o(1)}$ theorem; and

(d) subtracting a "gain" already built into an upstream packet spends the same cancellation twice.

Proof. The first statement is the disjoint union of event identifiers. Items (a)–(c) violate that union or replace missing information by a new estimate. For (d), the upstream exponent already contains the gain, so subtracting it again changes the proved bound. □

Remark 2.4 (Composition can improve the sum)*.* The product estimate $\|T_k \cdots T_1\| \le \prod_i \|T_i\|$ adds individual exponents. A direct bound for the literal composition may be smaller. It is legitimate precisely when recorded as a joint token replacing, not supplementing, the individual tokens.

3 The current physical registry

The canonical names below implement Hypothesis H9 of TPC-MVP1 [1]. They are not assertions that the corresponding exponents are known.

Token	Literal event	Amplitude conversion	Present status
λ_{frame}	TPC-113 canonical physical synthesis	Exponent of the actual H9 composite norm. For a matching $\mathcal{S}Q\mathcal{S}^\dagger$ factorization, $\kappa_{\mathrm{q}}\|Q\|$ is only a uniform upper certificate; a route stop requires an actual composite lower certificate	Growing operator factorization and norm cost unknown.
λ_{group}	TPC-114 residual grouping/native forgetting	Derived from the actual H9 norm; in ℓ^2, the amplitude norm is $\sqrt{k_{\max}}$ and the energy factor is $k_{\max}$	Actual growing fiber census and max-packet-to-scalar conversion not certified.
λ_{loc}	TPC-115 evaluation at fixed h_0	$\frac{1}{2}\log_X K_{h_0}$ from energy recovery	Actual physical profile regularity unknown.
λ_{tail}	TPC-116 complete high/ultra-tail return	Literal aggregate amplitude exponent	Complete outer bound unknown.
λ_{cover}	TPC-117 TPC-18 cover/residual transport	Canonical coefficient or residual cost	Growing matrix/vector certificate absent.
λ_{bdry}	Remaining boundary, mask and normalization transport	Direct amplitude exponent	Must be identified, not hidden in $o(1)$.

If, for example, the TPC-114 native-Gram event is already part of a direct joint Gram theorem with the TPC-113 frame, one replaces the two corresponding tokens by that proved joint token. Without this literal factorization they remain distinct. Numerically similar costs are not automatically duplicates.

4 Routewise linear programming

Scale parameters such as frequency cutoffs and block lengths often enter the exponent bounds piecewise affinely. Split the allowed region into finitely many route/cell choices. On one cell, introduce $x \in \mathbb{R}^n_{\geq 0}$ and write all proved scale restrictions as

$$Ax \geq b.$$

After the registry has removed duplicates, its loss is

$$\Lambda_{\text{phys}}(x) = c^{\mathsf{T}} x + d.$$

All entries should be exact rationals when a strict endpoint is certified; decimal approximations cannot decide equality.

Theorem 4.1 (Exact primal–dual endpoint certificates)**.** *For one fixed route and one complete actual theorem-backed registry, let every constraint and objective coefficient below refer to that same registry. For the primal problem*

$$\Lambda_* = \inf\{c^{\mathsf{T}} x + d : Ax \geq b,\ x \geq 0\},$$

consider the dual feasible set

$$y \geq 0, \qquad A^{\mathsf{T}} y \leq c,$$

with objective $b^{\mathsf{T}} y + d$. Then:

(i) every primal feasible x with $c^{\mathsf{T}} x + d < \frac{1}{400}$ certifies strict endpoint compatibility for that complete theorem-backed registry;

(ii) every dual feasible y with $b^{\mathsf{T}} y + d \geq \frac{1}{400}$ proves that this route cannot pass the strict endpoint;

(iii) if exact rational primal and dual points have the same objective, that value is Λ_*; in particular, a common value $\frac{1}{400}$ is an exact equality stop.

Proof. For primal feasible x and dual feasible y,

$$b^{\mathsf{T}} y \leq (Ax)^{\mathsf{T}} y = x^{\mathsf{T}} A^{\mathsf{T}} y \leq x^{\mathsf{T}} c.$$

This is weak duality. It proves (i) directly and (ii) by comparing every primal point with the dual lower bound. Equality of one primal and one dual objective traps the infimum between the same number, proving (iii). □

Remark 4.2 (A feasible scale point is not an arithmetic theorem)*.* Item (i) is callable only when every registry token is backed by the theorem whose hypotheses define the LP constraints. Filling an unknown arithmetic exponent with an optimizer variable and allowing the LP to choose it freely is not a proof.

Corollary 4.3 (Finite route menu)**.** *If a program has finitely many literal routes, it passes the endpoint when at least one complete route has a theorem-backed primal certificate below* $1/400$*. A route is stopped only by a dual certificate at or above* $1/400$ *for that same complete, actual, theorem-backed registry. If every route is stopped, a new representation or genuinely stronger arithmetic input is required.*

5 Equality, uncertainty and rerouting

Proposition 5.1 (Total three-valued endpoint audit)**.** *For a declared route, assign the first applicable certified state:*

(i) ***GO****: every token is theorem-backed and an exact upper certificate gives* $\Lambda_{\text{phys}} < 1/400$*;*

(ii) ***STOP ROUTE****: for the same complete actual registry, an exact lower/dual certificate gives* $\Lambda_* \geq 1/400$*;*

(iii) ***INCOMPLETE****: neither certificate is available. This includes an unknown token or constraint, unresolved primal feasibility or boundedness, an incomplete registry, or simply the absence of matching certified upper/lower information.*

These rules assign exactly one audit label: weak duality makes GO and STOP ROUTE incompatible, and INCOMPLETE contains every remaining case. The state INCOMPLETE may not be silently promoted to GO by setting unknown exponents to zero.

Proof. The first two alternatives are theorem 4.1. A strict primal upper bound below $1/400$ and a dual lower bound at or above $1/400$ cannot coexist by weak duality. If neither certified alternative holds, the definition assigns INCOMPLETE, irrespective of why the certification is missing. □

At a STOP ROUTE certificate, the permitted reroutes are concrete:

1. prove a direct joint operator theorem and replace costly constituent tokens;
2. choose a different lossless physical frame or full-block carrier;
3. retain signed cancellation before a positive majorant;
4. prove the TPC-117 physical residual small without exact cover; or
5. record the obstruction as a negative theorem and move to a different branch.

None of these operations changes the endpoint inequality; each changes the theorem-backed registry.

6 Exact rational regression certificate

The standard-library script `experiments/tpc118_endpoint_certificate.py` verifies:

1. deterministic merging only of identical full physical-event keys, nonmerging of distinct maps, and rejection of conflicting or alternative-source records for one event;
2. exact replacement of declared dependencies by a joint token, with unresolved dependencies rejected;
3. exact rational primal and dual feasibility;
4. matching optima strictly below, exactly equal to and strictly above $1/400$;
5. weak duality in each certificate; and
6. refusal to declare GO when one required token is unknown.

The three small LPs are certificate-format regressions, not fitted exponents for the actual TPC carrier.

7 Current verdict and claim boundary

Status	Exact requirement	TPC consequence
Go	Complete theorem-backed registry and rational primal value $< 1/400$	A strict physical margin exists; proceed to final synthesis without reusing any token.
Equality stop	Matching primal/dual value $= 1/400$	No fixed-power margin remains; this route does not pass.
Above-endpoint stop	Dual value $> 1/400$	The current route is quantitatively impossible; reroute.
Current state	Several growing physical exponents in the registry are unknown	INCOMPLETE, not GO and not a negative theorem for twin primes.

Established. The canonical registry logic, weak-duality endpoint test and exact equality stop are L0 results. The mapping from TPC-113–TPC-117 interfaces to nonduplicated physical tokens is an L1 audit.

Not established. No complete actual-carrier token table with proved growing exponents is supplied, so no strict $\Lambda_{\text{phys}} < 1/400$ certificate is claimed. There is no new L2 fixed-h_0 estimate, determinant-energy gain, parity breakthrough, Hardy–Littlewood asymptotic, prime-pair lower bound or twin-prime theorem. The next productive step is to measure or bound the unknown tokens one by one, beginning with the largest certifiable lower/upper interval, and to rerun the exact primal–dual audit after each theorem.

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